



Principles of Finance

Chapter 9

End-of-Chapter Questions

Time Value of Money III: Unequal Multiple Payment Values

Multiple Choice Questions

General instructions: Approximations and minor differences because of rounding are acceptable. Ignore the effect of taxes. Please assume that all percentages are annual rates and compounding occurs annually unless otherwise indicated.

1. For which of the following situations would you NOT be able to use the one-step shortcut provided by an annuity calculation?
 - A. Planning an amortization table for a three-year car loan at 2%
 - B. Investing \$2,000 per year for 10 years at 5%
 - C. Calculating the monthly payment on a 20-year home mortgage of \$150,000 at 3%
 - D. Finding the present value of a fund that will pay \$1,000 the first year, \$2,000 in the second year, and \$3,500 in the third year

Solution: D. The first three answers all involve an even stream of payments and can be solved with an annuity calculation. Answer D is a mixed stream of receipts and therefore must be solved either with three separate calculations plus a sum or with a tool such as Excel or a financial calculator.

Section 9.1 LO 1 Easy

2. If you invest \$3,000 today, \$5,000 one year from now, and \$3,000 two years from now, approximately how much money will you have at the end of the third year if you invest in a fund paying 7%?
 - A. \$11,770.24
 - B. \$12,609.63
 - C. \$13,187.79
 - D. \$14,274.50

Solution: B. The first \$3,000 is invested for three years, the \$5,000 for two years, and the second \$3,000 for one year. Using the formula $FV = PV \times (1 + i)^n$, we can calculate the future value of each invested amount and then combine them:

$$FV_1 = \$3,000 \times (1 + 0.07)^3 \approx 3,675.13$$

$$FV_2 = \$5,000 \times (1 + 0.07)^2 = 5,724.50$$

$$FV_3 = \$3,000 \times (1 + 0.07)^1 = 3,210.00$$

$$FV_1 + FV_2 + FV_3 = 3,675.13 + 5,724.50 + 3,210.00 = \$12,609.63$$

Section 9.1 LO 2 Easy

3. You intend to invest four annual payments, starting immediately: \$10,000 now, another \$10,000 at the end of year 1, \$12,000 at the end of year 2, and \$15,000 at the end of year 3. How much will you have at the end of year 4 if the fund is always earning 6% compounded annually?
- A. \$53,918.13
 - B. \$54,417.52
 - C. \$57,685.30
 - D. \$57,894.77

Solution: A. The first \$10,000 is invested for four years, the second \$10,000 for three years, the \$12,000 for two years, and the \$15,000 for one year. Using the formula $FV = PV \times (1 + i)^n$, we can calculate and combine:

$$FV_1 = \$10,000 \times (1 + 0.06)^4 \approx 12,624.77$$

$$FV_2 = \$10,000 \times (1 + 0.06)^3 = 11,910.16$$

$$FV_3 = \$12,000 \times (1 + 0.06)^2 = 13,483.20$$

$$FV_4 = \$15,000 \times (1 + 0.06)^1 = 15,900.00$$

$$12,624.77 + 11,910.16 + 13,483.20 + 15,900.00 = \$53,918.13$$

Section 9.1 LO 2 Easy

4. You can invest a windfall of \$40,000 today at 4% compounded annually. You don't believe you can make another investment one year from now, but you believe that you can save enough that two years from now, you can deposit another \$10,000 in the same investment, which will still be earning 4% compounded annually. How much will you have accumulated in this investment three years from now?
- A. \$50,400.00
 - B. \$54,875.32
 - C. \$55,394.56
 - D. \$57,123.33

Solution: C. Your first investment of \$40,000 will compound for three years, with nothing added at the start of year 2 but with another \$10,000 added at the end of year 2 that will compound for one more year. The calculations are as follows:

$$FV_1 = \$40,000 \times (1 + 0.04)^3 = \$44,994.56$$

$$FV_2 = \$10,000 \times (1 + 0.04)^1 = \$10,400.00$$

$$\$44,994.56 + \$10,400.00 = \$55,394.56$$

Section 9.1 LO 2 Easy

5. Donna can invest today in a fund that will guarantee her a 5% return compounded annually for five years. She can invest \$1,200 today, \$1,400 one year from now, \$1,800 two years from now, and \$2,100 three years from now. If she can make these investments as scheduled, how much will she have accumulated five years from now?
- A. \$6,878.97
 - B. \$7,632.23
 - C. \$8,142.52
 - D. \$8,799.49

Solution: B. Because Donna's money will work for her on a five-year schedule, noting that there is no investment made at the start of year 5, we frame the timing and earning power of each investment as follows:

$$FV_1 = \$1,200 \times (1 + 0.05)^5 \approx 1,531.54$$

$$FV_2 = \$1,400 \times (1 + 0.05)^4 \approx 1,701.71$$

$$FV_3 = \$1,800 \times (1 + 0.05)^3 \approx 2,083.73$$

$$FV_4 = \$2,100 \times (1 + 0.05)^2 = 2,315.25$$

$$1,531.54 + 1,701.71 + 2,083.73 + 2,315.25 = \$7,632.23$$

Section 9.1 LO 2 Easy

6. Jane wants to create a fund today that will provide her a 4% guaranteed compounded annual rate. She wants to withdraw \$15,000 one year from now and \$27,000 two years from now, after which the fund will be depleted. How much must she invest today to achieve this goal?
- A. \$42,400.00
 - B. \$41,912.43
 - C. \$40,775.89
 - D. \$39,386.10

Solution: D. Each of the withdrawal amounts must be discounted separately to today because they do not represent an annuity. We remember that the formula for the present value of a single amount is $PV = FV \times \frac{1}{(1+i)^n}$.

We calculate the present value of each of the amounts in the problem, then combine them:

$$PV_1 = \$15,000 \times \frac{1}{(1 + 0.04)^1} \approx 14,423.08$$

$$PV_2 = \$27,000 \times \frac{1}{(1 + 0.04)^2} \approx 24,963.02$$

$$PV_1 + PV_2 = 14,423.08 + 24,963.02 = \$39,386.10$$

Section 9.1 LO 3 Easy

7. You wish to draw from a fund you're creating today with a 3% guaranteed compounded annual rate. You hope to withdraw \$40,000 one year from now, \$35,000 two years from now, and \$25,000 three years from now, at which point the fund will be depleted. How much must you invest today to achieve this?
- A. \$94,704.35
 - B. \$95,346.98
 - C. \$95,255.12
 - D. \$95,945.78

Solution: A.

$$PV_1 = \$40,000 \times \frac{1}{(1 + 0.03)^1} \approx 38,834.95$$

$$PV_2 = \$35,000 \times \frac{1}{(1 + 0.03)^2} \approx 32,990.86$$

$$PV_3 = \$25,000 \times \frac{1}{(1 + 0.03)^3} \approx 22,878.54$$

$$38,834.95 + 32,990.86 + 22,878.54 = \$94,704.35$$

Section 9.1 LO 3 Medium

Review Questions

1. Refer to Problem 3 above. Solve using the financial calculator, documenting your steps/keystrokes.

Section 9.1 LO 2 Easy

2. Refer to Problem 7 above. Solve using the financial calculator, documenting your steps/keystrokes.

Section 9.1 LO 3 Medium

3. You agree to repay a loan over five years with the following stream of cash payments: \$1,000; \$1,100; \$1,250; \$1,280; and \$1,300. If you wish to discount these payments to their present value today using 4%, why can you not use one annuity calculation, as seen in previous chapters?

Solution: Annuity techniques can only be used when all amounts in a stream of payments or receipts are equal. Because the amounts of these payments vary, an annuity formula cannot be used. Each payment must be individually discounted at 4% to present value, and then the five results can be combined to find the total present value of this mixed cash stream.

Section 9.1 LO 1 Easy

Problems

1. Your aunt promises to gift you \$1,500 now, \$1,700 one year from now, \$1,900 two years from now, and \$2,500 three years from now. You will deposit all four amounts in an account that bears 3% interest compounded annually. How much will be in the account at the end of the fourth year?

Solution: We calculate the future value of each amount in this stream of payments and combine them, using the formula $FV = PV \times (1 + i)^n$ for each:

$$FV_1 = \$1,500 \times (1 + .03)^4 \approx 1,688.26$$

$$FV_2 = \$1,700 \times (1 + .03)^3 \approx 1,857.64$$

$$FV_3 = \$1,900 \times (1 + .03)^2 = 2,015.71$$

$$FV_4 = \$2,500 \times (1 + .03)^1 = 2,575.00$$

$$1,688.26 + 1,857.64 + 2,015.71 + 2,575.00 = \$8,136.61$$

Section 9.1 LO 2 Easy

2. You believe that you can set aside \$1,200 each year for the next four years, starting immediately, in order to buy a small fishing boat for your retirement. Your friend Luis promises that he'll pay you back \$4,900 that he owes you three years from now, so you will add that to the payment you make at the start of year 4. Then, at the start of year 5, you will increase your payment to \$1,400; in year 6, to \$1,500; and in year 7, to \$1,600. Every payment will be deposited in a fund bearing 4% interest compounded annually. How much will you have set aside for your boat at the end of the seventh year?

Solution: We could perform seven separate calculations on each annual payment, but notice that the first three years actually represent an annuity, as learned in previous chapters. Because we begin making payments immediately, it's technically an annuity due. We can approach this problem by finding the value of that four-year annuity and then adding each of the future value calculations for the three following years. The formula for the future value of an annuity due is

$$FVa = PYMT \times \frac{(1 + i)^n - 1}{i} \times (1 + i)$$

We will only use this formula for the first three years. Recall that each annuity payment is \$1,200:

$$FVa = \$1,200 \times \frac{(1 + 0.04)^3 - 1}{0.04} \times (1 + 0.04)$$

$$FVa = \$1,200 \times 3.1216 \times (1 + 0.04) = \$3,895.76$$

The student is reminded that there are other approaches to finding the solution. For the moment, we will take the approach of also growing this number for four more years, in addition to calculating the future values of the four subsequent payment amounts. We can't forget that this number, now three years in our future, will continue to work for you. We will begin by growing this future value of the annuity due for four more years as a single amount:

$$FV_1 = \$3,895.76 \times (1 + 0.04)^4 \approx \$4,557.49$$

Next, we find the future value of each following amounts, keeping in mind how many years n each amount has to compound:

$$FV_2 = (\$1,200 + \$4,900) \times (1 + 0.04)^4 \approx \$7,136.14$$

$$FV_3 = \$1,400 \times (1 + 0.04)^3 \approx \$1,574.81$$

$$FV_4 = \$1,500 \times (1 + 0.04)^2 = \$1,622.40$$

$$FV_5 = \$1,600 \times (1 + 0.04)^1 = \$1,664.00$$

Combining all five amounts produces the following sum:

$$4,557.49 + 7,136.14 + 1,574.81 + 1,622.40 + 1,664.00 = \$16,554.84$$

Section 9.1 LO 2 Medium

3. Assume you win a lottery that will pay you \$10,000 immediately, plus \$20,000 one year from now, \$30,000 two years from now, \$40,000 three years from now, and \$75,000 four years from now. You're curious about how much the lottery commission might offer you as an immediate lump sum instead. What is the minimum amount you should consider accepting today instead? Use a 5% annual discount rate. Please remember that we are ignoring taxation and other considerations and basing this only on the math itself.

Solution: We are trying to find today's equivalent of uneven payments. \$10,000 is already at its present value today; however, we must take the other four future payments and discount them to today at 5%. We then add all five present values, and that total is the cash equivalent you should expect, all other things being equal. If you are offered any amount less than that, you should opt for the future payment stream instead. Using the present value formula $PV = FV \times \frac{1}{(1+i)^n}$, we discount each of the future payments to today and then combine them with the \$10,000 immediate payment (which is already at its present value):

$$PV_1 = \$10,000$$

$$PV_2 = \$20,000 \times \frac{1}{(1 + 0.05)^1} \approx \$19,047.62$$

$$PV_3 = \$30,000 \times \frac{1}{(1 + 0.05)^2} \approx \$27,210.88$$

$$PV_4 = \$40,000 \times \frac{1}{(1 + 0.05)^3} \approx \$34,553.50$$

$$PV_5 = \$75,000 \times \frac{1}{(1 + 0.05)^4} \approx \$61,702.69$$

$$\$10,000 + \$19,047.62 + \$27,210.88 + \$34,553.50 + \$61,702.69 = \$152,514.69$$

All things being equal, you should not accept a cash lump sum lower than \$152,514.69.

Section 9.1 LO 3 Medium

4. Assume you win a lottery, and you are offered the following stream of payments by the lottery commission: \$25,000 today, \$32,000 one year from now, another \$32,000 two years from now, and a final payment of \$55,000 three years from now. You accept the offer. If you invest all of these proceeds at 6% compounded annually and extract nothing from the investment, how much will you have at the end of the fourth year?

Solution: We calculate the future value of each payment using our formula for the future value of a single amount, $FV = PV \times (1 + i)^n$:

$$FV_1 = \$25,000 \times (1 + 0.06)^4 \approx \$31,561.92$$

$$FV_2 = \$32,000 \times (1 + 0.06)^3 \approx \$38,112.51$$

$$FV_3 = \$32,000 \times (1 + 0.06)^2 = \$35,955.20$$

$$FV_4 = \$55,000 \times (1 + 0.06)^1 = \$58,300.00$$

$$\$31,561.92 + \$38,112.51 + \$35,955.20 + \$58,300.00 = \$163,929.63$$

Section 9.1 LO 2 Medium

5. You are offered a business partnership that guarantees you cash returns of \$150,000 one year from now, nothing at the end of year 2, and \$350,000 at the end of year 3. After year 3, the partnership will be dissolved, and there will be no more expected returns on your investment. If you analyze this plan expecting 7% compounded annually, what is this potential deal worth to you today?

Solution: This is solved by finding the present value of each expected amount in the future stream, discounted at 7%, then combining them. No amount is used for year 2 because there is no expectation of a cash inflow for that period.

$$PV_{\text{Year 1}} = \$150,000 \times \frac{1}{(1 + 0.07)^1} \approx \$140,186.92$$

$$PV_{\text{Year 3}} = \$350,000 \times \frac{1}{(1 + 0.07)^3} \approx \$285,704.26$$

$$140,186.92 + 285,704.26 = \$425,891.18$$

Section 9.1 LO 3 Medium

6. Tony owns a small business that he is attempting to sell. A potential buyer offers him \$500,000 today, plus \$1,500,000 two years from now and a balance of \$1,700,000 three years from now. Tony always analyzes cash flows using a rate of 4% compounded annually. To compare this to other offers, which would be paid in cash immediately, he wants to know the present value of these future cash flows. Show the calculation and solution that you would present to Tony to use in his decision.

Solution: No discounting is needed for the immediate cash payment. Using the PV formula, the other two future amounts are discounted to today at 4% and then combined with the cash payment for the total present value:

$$\begin{aligned}
 PV_1 &= \$500,000 \\
 PV_2 &= \$1,500,000 \times \frac{1}{(1 + 0.04)^2} \approx \$1,386,834.32 \\
 PV_3 &= \$1,700,000 \times \frac{1}{(1 + 0.04)^3} \approx \$1,511,293.81 \\
 \$500,000 + \$1,386,834.32 + \$1,511,293.81 &= \$3,398,128.13
 \end{aligned}$$

All things being equal, Tony should not accept a cash lump sum lower than \$3,398,128.13.

Section 9.1 LO 3 Medium

7. Continuing Problem 6 above, assume that as Tony receives each payment from the buyer, he can immediately invest it in a fund that returns a guaranteed 3.5% compounded annually. If he accepts the terms of this offer, how much will he have accumulated in this investment four years from now?

Solution: We begin investing immediately and calculate the future value of each single amount as it's added to Tony's investment, accumulating interest at 3.5% throughout. The student is reminded to draw a timeline to assist with the solution. Note that no amount is added at the end of year 1.

$$\begin{aligned}
 FV_1 &= \$500,000 \times (1 + 0.035)^4 = \$573,761.50 \\
 FV_2 &= \$1,500,000 \times (1 + 0.035)^2 = \$1,606,837.50 \\
 FV_3 &= \$1,700,000 \times (1 + 0.035)^1 = \$1,759,500.00 \\
 \$573,761.50 + \$1,606,837.50 + \$1,759,500.00 &= \$3,940,099
 \end{aligned}$$

Section 9.1 LO 2 Difficult

8. You hire Susan for a three-year term. She has no retirement plan, so you agree to invest money immediately to allow her a stream of three payments beginning one year after her employment term ends. Draw a timeline! The money you invest for the full six years of this arrangement (three years of employment and three years of withdrawals) always earns 4% compounded annually. When Susan receives her third and last payment, the fund will be depleted and equal zero. The three payments she will receive are as follows:

End of year 4: \$25,000
 End of year 5: \$30,000
 End of year 6: \$37,000

Therefore, your goal is to have enough money in this account at the end of Susan's three-year employment term to assure her of receiving these payments.

How much money must you invest today to accomplish this strategy?

Solution: This is a two-stage problem. First, we must find the present value of Susan's expected cash flows, but this "present" value will be at a time three years in the future. Once we determine the amount needed to fund her three payments, we discount that single amount to today to find how much must be invested now for the account to have sufficient funds on the last day of year 3, when her employment ends and her three years of withdrawals begin.

First, we discount the three years of payments Susan will receive (at 4%) down to the starting amount of the account on the last day of year 3:

$$PV_1 = \$25,000 \times \frac{1}{(1 + .04)^1} \approx \$24,038.46$$

$$PV_2 = \$30,000 \times \frac{1}{(1 + .04)^2} \approx \$27,736.69$$

$$PV_3 = \$37,000 \times \frac{1}{(1 + .04)^3} \approx \$32,892.87$$

$$\$24,038.46 + \$27,736.69 + \$32,892.87 = \$84,668.02$$

This is the amount we wish to have available at the end of Susan's employment, the last day of year 3. One year later, on the last day of year 4, she will receive her first payment of \$25,000. This amount will be enough to fund her three annual withdrawals, after which the fund will be depleted.

The question, however, is the amount needed today. This single amount will grow to the \$84,668.02 goal that must be met three years from now. This amount will also compounded annually at 4%. Using techniques from preceding chapters to discount the goal amount, this amount that must be invested today is

$$PV = \$84,668.02 \times \frac{1}{(1 + 0.04)^3} \approx \$75,269.56$$

This is our solution. The student might be misled by performing calculations to find a "present" value three years in the future but is reminded that when that calculation of the present value of an uneven stream is performed, we are looking at it as if we were three years in the future.

Section 9.1 LO 3 Difficult

9. Abby owns a business that projects two years of strong cash flow, followed by a two-year hiatus while she pursues a graduate degree. She then plans to resume her business and is confident about the following two years of cash flow. At the end of each year, she will invest her profits in a fund that returns 3% compounded annually. Each investment will be made at the end of the year. Her expected investments are as follows:

Year 1: \$50,000

Year 2: \$78,000

Year 3: 0

Year 4: 0

Year 5: \$84,000

Year 6: \$89,000

If Abby meets her expectations, how much money will she have in this fund at the end of year 7?

Solution: In year 1, there are no interest earnings because the funds are invested at the end of the year. Because we're looking for the amount at the end of seven years, Abby's year 1 investment will compound for six years, and so on. The student must pay careful attention to the timing of each investment:

$$FV_1 = \$50,000 \times (1 + 0.03)^6 \approx \$59,702.61$$

$$FV_2 = \$78,000 \times (1 + 0.03)^5 \approx \$90,423.38$$

$$FV_3 = \$84,000 \times (1 + 0.03)^2 = \$89,115.60$$

$$FV_4 = \$89,000 \times (1 + 0.03)^1 = \$91,670.00$$

$$\$59,702.61 + \$90,423.38 + \$89,115.60 + \$91,670.00 = \$330,911.59$$

Section 9.1 LO 2 Difficult

Chapter 9

Test Bank

Time Value of Money III: Unequal Multiple Payment Values

These questions and problems can all be solved in a classroom setting using a handheld financial calculator.

True or False Questions

1. Problems involving multiple unequal payments require that you analyze the payments as individual separate payments.

- A. True
- B. False

Solution: A. True

Section 9.1 LO 1 Easy

2. In calculating the future value of a stream of multiple unequal payments, raising the interest rate will result in the sum of the future values rising.

- A. True
- B. False

Solution: A. True

Section 9.1 LO 2 Moderate

3. You cannot use annuity methods in analyzing multiple unequal payments.

- A. True
- B. False

Solution: A. True

Section 9.1 LO 2 Easy

4. In determining the present value of a stream of multiple unequal payments, raising the discount rate will result in the sum of the present values rising.

- A. True
- B. False

Solution: B. False

Section 9.1 LO 3 Moderate

5. Doubling the individual payments in a stream of multiple unequal payments will result in the sum of the present values doubling.

A. True
B. False

Solution: A. True

Section 9.1 LO 3 Moderate

Multiple Choice Questions

1. Investing is a risky business, so investors must be ready to accept that future investment cash flows may be uncertain and unpredictable. This being case, what is the best way to evaluate the value of such an investment?

A. Determine the future value of the individual anticipated cash flows at a minimum acceptable rate of return
B. Convert the individual future cash flows in a perpetuity and determine the present value of the perpetuity
C. Convert the individual future cash flows into an annuity and determine the present value of the annuity
D. Determine the present value of the individual anticipated cash flows at a minimum acceptable rate of return

Solution: D. Enter the sequence of cash flows using the CF function on a financial calculator or enter the cash flows as a range in the Excel PV function. Enter the minimum acceptable rate of return as the interest rate. Solve for present value. The answer will be calculated last.

Section 9.1 LO 1 Easy

2. An investor has developed projections for an investment that will generate the following stream of future cash flows: Yr1 = 50, Yr2 = 60, Yr3 = 75, Yr4 = 80, Yr5 = 90. She believes that she can invest her money at 10 percent annually. How much will she have accumulated by the fifth year of this investment?

A. 421.82
B. 363.21
C. 287.82
D. 315.91

Solution: A. Enter the sequence of cash flows using the CF function on a financial calculator or enter the cash flows as a range in the Excel PV function. Enter 10 percent as the interest rate. Solve for future value. The answer will be calculated last.

Section 9.1 LO 2 Moderate

3. Given the following stream of future cash flows and an interest rate of 8 percent, what is its future value? Yr1 = 50, Yr2 = 60, Yr3 = 75, Yr4 = 80, Yr5 = 90.

A. 351.58
B. 422.49
C. 407.49
D. 273.48

Solution: C. Enter the sequence of cash flows using the CF function on a financial calculator or enter the cash flows as a range in the Excel PV function. Enter 8 percent as the interest rate. Solve for future value. The answer will be calculated last.

Section 9.1 LO 2 Moderate

4. A farmer has developed projections of future cash flows from a new variety of grain. Yr1 = 50, Yr2 = 60, Yr3 = 75, Yr4 = 80, Yr5 = 90. He has a minimum required rate of return on his investments of 10 percent. How much should he be willing to pay for the investment opportunity?

A. 290
B. 240
C. 315
D. 262

Solution: D. Enter the sequence of cash flows using the CF function on a financial calculator or enter the cash flows as a range in the Excel PV function. Enter 10 percent as the interest rate. Solve for present value. The answer will be calculated last.

Section 9.1 LO 3 Moderate

5. Given the following stream of future cash flows and an interest rate of 8 percent, what is its present value? Yr1 = 50, Yr2 = 60, Yr3 = 75, Yr4 = 80, Yr5 = 90.

A. 355
B. 417
C. 277
D. 312

Solution: C. Enter the sequence of cash flows using the CF function on a financial calculator or enter the cash flows as a range in the Excel PV function. Enter 8 percent as the interest rate. Solve for present value. The answer will be calculated last.

Section 9.1 LO 3 Moderate

6. An investor is evaluating a “boom or bust” investment that in some years will generate an excellent cash flow and in some years will generate no cash at all or a negative cash flow. The following projected cash flow reflects this: Yr1 = 100, Yr2 = negative 25, Yr3 = 150, Yr4 = 0, Yr5 = 130. If her minimum required rate of return is 10 percent, what is the maximum amount she should pay for this investment?
- A. 350
 - B. 530
 - C. 263
 - D. 425

Solution: C. Enter the sequence of cash flows using the CF function on a financial calculator. Enter 10 percent as the interest rate. Solve for present value. The answer will be calculated last.

Section 9.2 LO 1 Medium

7. An investor is evaluating a “boom or bust” investment that in some years will generate an excellent cash flow and in some years will generate no cash at all or a negative cash flow. The following projected cash flow reflects this: Yr1 = 100, Yr2 = negative 25, Yr3 = 150, Yr4 = 0, Yr5 = 130. If her minimum required rate of return is 10 percent, what will be the value of her investment at the end of the investment horizon?
- A. 350
 - B. 530
 - C. 370
 - D. 425

Solution: D. Enter the sequence of cash flows as range in the Excel FV function. Enter 10 percent as the interest rate. Solve for future value. The answer will be calculated last.

Section 9.2 LO 2 Difficult

Problems

1. An investor has to decide how much he will be willing to pay for an investment that generates the following stream of future cash flows: Yr1 = 35, Yr2 = 40, Yr3 = 45, Yr4 = 50, Yr5 = 55. His minimum required rate of return is 9 percent. How much should he be willing to pay for it today?

Solution: 172. Enter the sequence of cash flows using the CF function on a financial calculator or enter the cash flows as a range in the Excel PV function. Enter 9 percent as the interest rate. Solve for present value. The answer will be calculated last.

Section 9.1 LO 3 Moderate

2. An investor has to decide how much he will be willing to pay for an investment that generates the following stream of future cash flows: Yr1 = 35, Yr2 = 40, Yr3 = 45, Yr4 = 50, Yr5 = 55. His minimum required rate of return is 11 percent. How much should he be willing to pay for it today?

Solution: 162. Enter the sequence of cash flows using the CF function on a financial calculator or enter the cash flows as a range in the Excel PV function. Enter 11 percent as the interest rate. Solve for present value. The answer will be calculated last.

Section 9.1 LO 3 Moderate

3. An investor is planning her retirement and has come across an investment that generates the following stream of future cash flows: Yr1 = 35, Yr2 = 40, Yr3 = 45, Yr4 = 50, Yr5 = 55. Her minimum required rate of return is 10 percent. How much should she plan on having by its fifth year?

Solution: 220. Enter the sequence of cash flows using the CF function on a financial calculator or enter the cash flows as a range in the Excel PV function. Enter 10 percent as the interest rate. Solve for future value. The answer will be calculated last.

Section 9.1 LO 2 Moderate

4. An investor is planning her retirement and has come across an investment that generates the following stream of future cash flows: Yr1 = 35, Yr2 = 40, Yr3 = 45, Yr4 = 50, Yr5 = 55. Her minimum required rate of return is 8 percent. How much should she plan on having by its fifth year?

Solution: 211. Enter the sequence of cash flows using the CF function on a financial calculator or enter the cash flows as a range in the Excel PV function. Enter 8 percent as the interest rate. Solve for future value. The answer will be calculated last.

Section 9.1 LO 2 Moderate